SentinelFlow

Real-Time DDoS Detection & Adaptive Mitigation

Product documentation report. SentinelFlow is a defensive network-monitoring application that detects suspicious / DDoS-like traffic with a Random Forest classifier, assesses threat severity, monitors network health, and applies adaptive mitigation with automatic recovery.

Measured model performance (this build)

Accuracy: 0.9971
Precision: 0.9996
Recall: 0.9946
F1-score: 0.9971
ROC-AUC: 0.9999
Test samples: 4800
Dataset: synthetic-fallback (CICIDS2017 unreachable; schema-compatible synthetic flows) + window-scale augment

1. SentinelFlow overview

SentinelFlow is a standalone defensive network-security product. It watches a selected network interface (or a safe local simulator), aggregates traffic into monitoring windows, classifies each window with a trained Random Forest, scores distributed-traffic risk, assigns an explainable severity, and triggers adaptive mitigation with automatic recovery.

2. Problem being addressed

volumetric and distributed denial-of-service traffic can overwhelm hosts and links. Operators need a lightweight, explainable, always-on monitor that flags suspicious windows within seconds, explains why they look suspicious, and takes proportionate defensive action without manual triage.

3. Key product features

- Real-time DDoS detection (Random Forest)
- 10-second rolling monitoring windows
- Distributed / carpet-bombing-aware risk scoring
- Explainable Low/Medium/High/Critical severity
- Adaptive mitigation: application-level quarantine with ~40 s auto-recovery
- Network health monitoring (CPU, memory, throughput)
- Simulation Mode for permission-free demos
- SQLite-backed detection, mitigation, and health history

4. User workflow

1. Launch SentinelFlow and open http://localhost:5000.
2. Pick a capture interface or press "Start Simulation".
3. Watch summary cards, health charts, and the detection table.
4. Click a row for threat detail and severity factors.
5. Review the Mitigation Center; quarantines recover automatically.

5. System architecture

Modular Python application: monitoring/ (Scapy capture, aggregation, simulator), ml/ (features, training, inference), security/ (detector, distributed analysis, severity, mitigation), database/ (SQLite), config/, Flask app.py serving server-rendered templates plus a JSON REST API, with a vanilla-JS dashboard using Chart.js (CDN with numeric fallback).

6. ML architecture

RandomForestClassifier (200 trees, class_weight=balanced, seed 42) over a compact 10-feature interface-level schema. No LLM is used for detection. Feature names and ordering are validated before every predict() call via the shared ml/features.py module.

7. CICIDS2017 data pipeline

The trainer first looks for local CICIDS2017 CSVs in data/raw/, then attempts documented public mirrors. In this build the dataset was unreachable, so a schema-compatible synthetic fallback was generated (clearly labelled in metadata). Raw flow fields are normalized, cleaned of infinite/missing values, and mapped forward->sent / backward->received with rates from flow duration. A documented window-scale augmentation

(6000 aggregated windows through the real monitoring pipeline) keeps the model calibrated across flow scale and window scale. Provenance: synthetic-fallback (CICIDS2017 unreachable; schema-compatible synthetic flows) + window-scale augmentation (6000 aggregated windows via monitoring pipeline)

8. Random Forest implementation

Stratified 80/20 train/test split, reproducible seed 42, class-balanced weights. Artifacts: persisted model (joblib), feature list, metadata, metrics JSON, classification report, confusion matrix, ROC curve, and feature importance.

9. Scapy monitoring

Scapy sniff() captures per-window packets on the chosen interface (default 10 s). Packets are grouped into received (inbound/local-destination) vs sent (outbound) using local interface addresses, then reduced to the 10 common features. Permission/driver failures surface clear errors and the UI offers Simulation Mode as a safe alternative.

10. Network Health Monitoring

psutil samples CPU, memory, and interface throughput into a rolling buffer plus SQLite history; the dashboard charts CPU/memory and inbound/outbound byte series.

11. Threat Severity Analysis

Severity blends model confidence, packet-rate pressure, distributed-risk score, source/destination spread, and cross-window persistence into Low/Medium/High/Critical, always listing the contributing factors next to the verdict.

12. Adaptive Mitigation

High/Critical windows quarantine the top suspicious sources inside SentinelFlow (default ~40 s, doubled for Critical) with automatic expiry and recovery history; Medium recommends rate limiting; Low keeps monitoring. OS firewall changes are never made automatically.

13. Simulation Mode

A safe local generator produces benign plus flood / carpet-bombing / slow-style windows, aggregates them through the identical inference pipeline, and labels the UI banner as Simulation Mode. It never emits network traffic.

14. Technology stack

Python 3.11/3.12, Flask, scikit-learn, pandas, numpy, joblib, Scapy, psutil, matplotlib, SQLite; HTML5 + modern CSS + vanilla JS + Chart.js frontend; Docker support.

15. How to run

Automatic: double-click run.bat and open http://localhost:5000. Docker: docker build -t sentinelflow . then docker run --rm -p 5000:5000 sentinelflow (or run-docker.bat). Manual: create a venv, pip install -r requirements.txt, python ml/train_model.py, python app.py. Full steps in HOW_TO_RUN.md.

16. Actual evaluation metrics (this build)

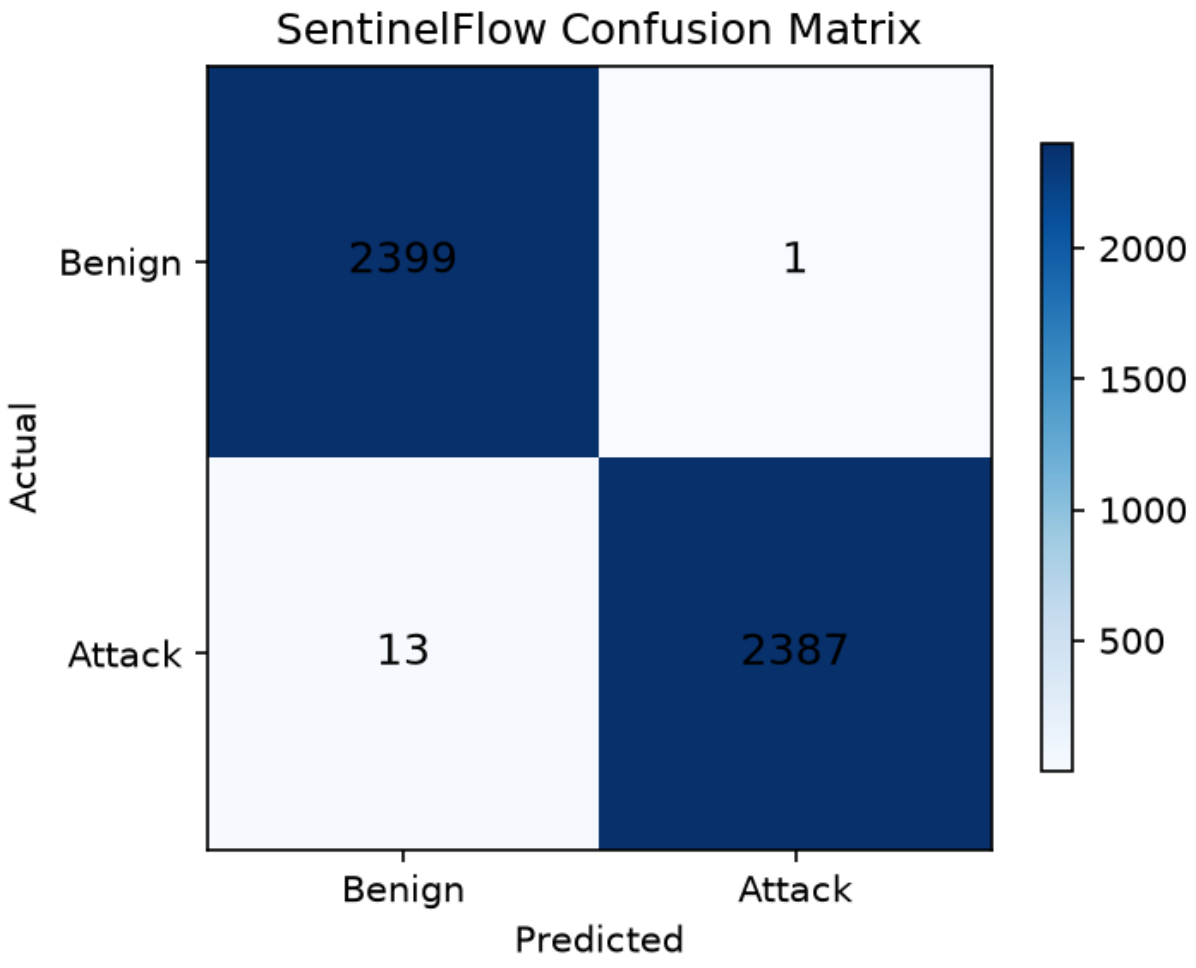
Accuracy: 0.9971
Precision: 0.9996
Recall: 0.9946
F1-score: 0.9971
ROC-AUC: 0.9999
Test samples: 4800 | Train samples: 19200
Class totals: benign=12000 attack=12000

Classification report:

	precision	recall	f1-score	support
Benign	0.99	1.00	1.00	2400
Attack	1.00	0.99	1.00	2400
accuracy			1.00	4800
macro avg	1.00	1.00	1.00	4800
weighted avg	1.00	1.00	1.00	4800

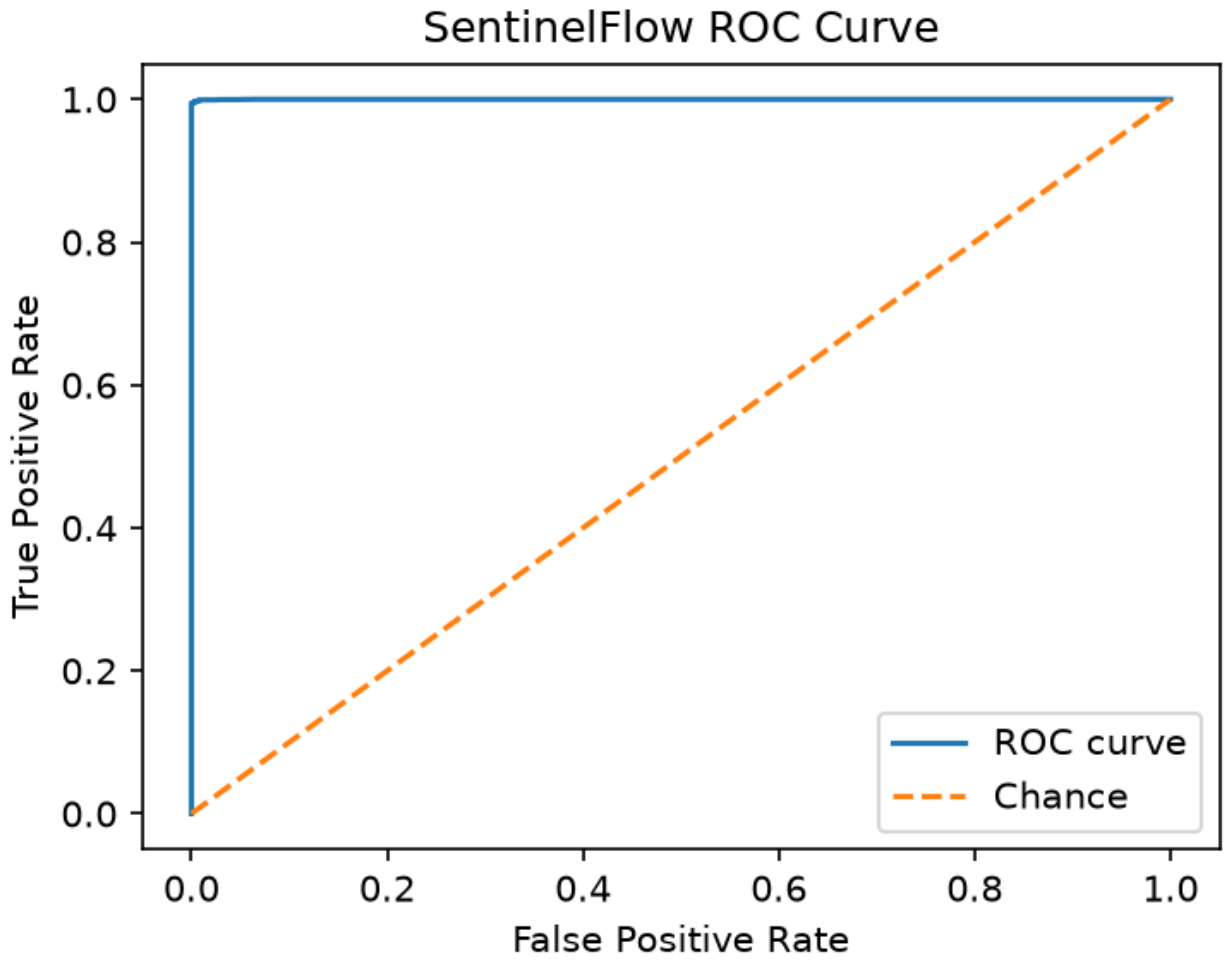
17. Actual confusion matrix

Rows = actual [Benign, Attack]; columns = predicted [Benign, Attack]: [[2399, 1], [13, 2387]]



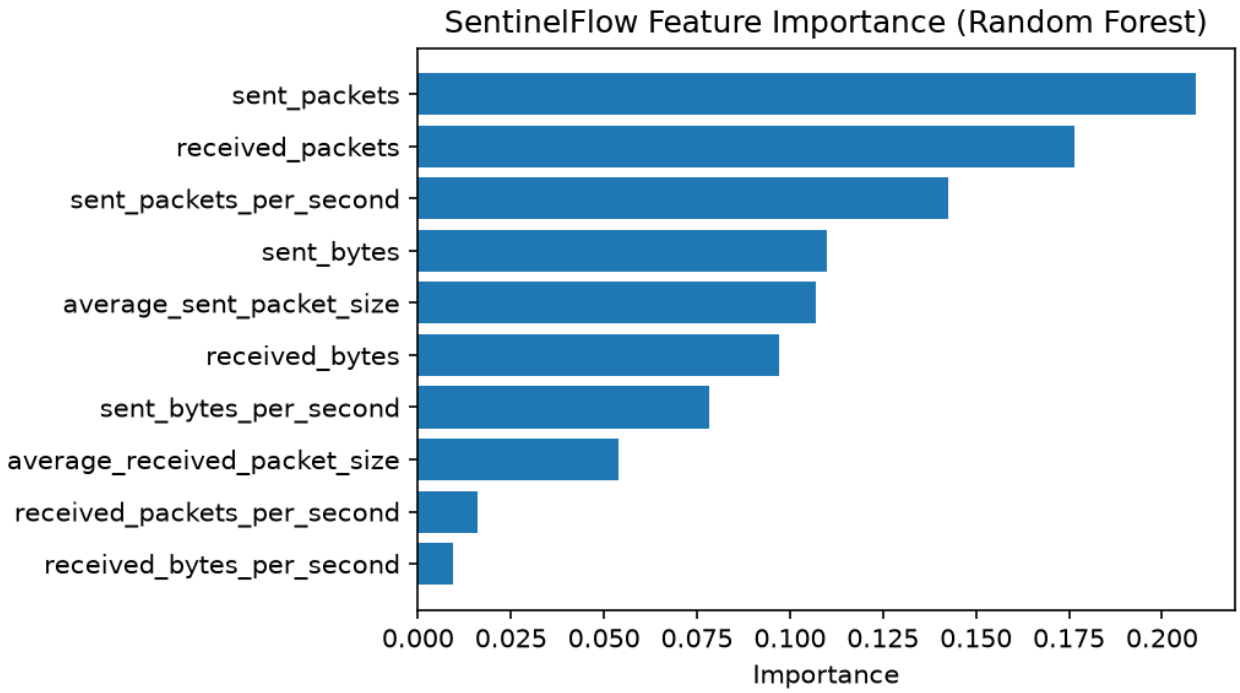
18. Actual ROC-AUC

ROC-AUC: 0.9999



19. Actual feature importance

- sent_packets: 0.2090
- received_packets: 0.1765
- sent_packets_per_second: 0.1425
- sent_bytes: 0.1100
- average_sent_packet_size: 0.1071
- received_bytes: 0.0970
- sent_bytes_per_second: 0.0783
- average_received_packet_size: 0.0540
- received_packets_per_second: 0.0160
- received_bytes_per_second: 0.0096



20. Limitations and responsible-use notes

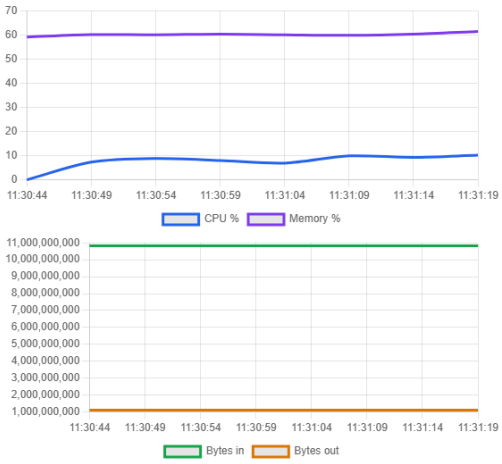
- Defensive monitoring only: no attack generation, no external scanning, no automatic firewall edits.
- Live capture needs elevated privileges and Npcap on Windows; otherwise use Simulation Mode.
- This build trained on documented synthetic fallback data (CICIDS2017 unreachable offline); metrics describe that data, not CICIDS2017 captures, and the dashboard labels this.
- Window aggregates differ in scale from single flows; window-scale augmentation addresses this but operators should re-train on their own labelled captures for production use.
- Binds to 127.0.0.1 only; localhost dashboard.

Appendix: SentinelFlow dashboard (running application)

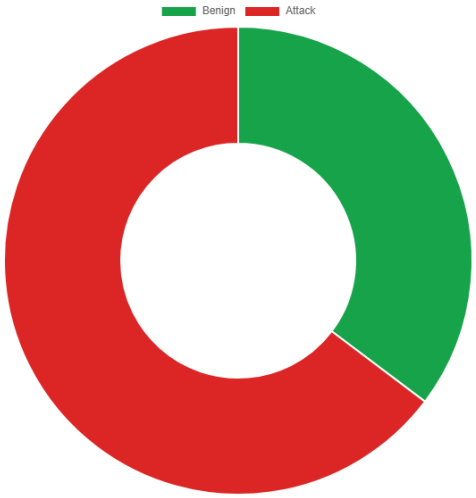
CURRENT THREAT STATUS	THREAT SEVERITY	ML CONFIDENCE	PACKETS / SEC	SOURCES / DESTINATIONS	ACTIVE MITIGATIONS
Attack	**High**	**100.0%**	**350.3**	**958**	**21**
simulation · 11:31:26	Distributed risk: 0.3784 (Low)	Random Forest	276,923.3 bytes/sec	dst 1	Mode: simulation

Network Health

CPU **10.2%** Memory **61.4%** In **10,822,878,222** Out **1,108,863,056**

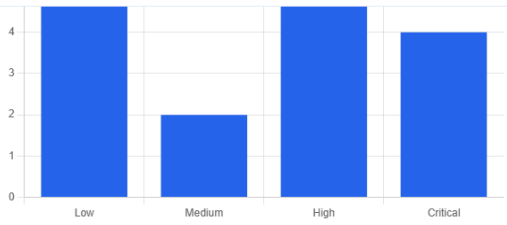


Detection Analytics



SentinelFlow
Real-Time DDoS Detection & Adaptive Mitigation

Simulation Mode | Bluetooth Network Connection (169.254.123.59, fe80::84a6:3763:a00d:e2a9) | Start Live | Start Simulation | Stop



Live Detection Table

Simulation Mode — synthetic input, real inference

Time	Interface	Source	Prediction	Conf.	Severity	Srcs	Dsts	Pkt/s	Mitigation
11:31:22	simulation	simulation	Attack	100.0%	Critical	37	8	1,592	quarantine
11:31:19	simulation	simulation	Benign	75.0%	Low	3	4	62	monitor
11:31:16	simulation	simulation	Benign	75.5%	Low	3	4	199	monitor
11:31:13	simulation	simulation	Attack	60.5%	Medium	3	4	89	rate_limit_recommended
11:31:10	simulation	simulation	Attack	60.5%	Medium	3	4	105	rate_limit_recommended
11:31:07	simulation	simulation	Attack	100.0%	Critical	29	1	2,902	quarantine
11:31:04	simulation	simulation	Attack	100.0%	High	559	1	199	quarantine
11:31:01	simulation	simulation	Benign	77.5%	Low	3	4	200	monitor
11:30:58	simulation	simulation	Benign	56.5%	Low	3	4	100	monitor
11:30:55	simulation	simulation	Attack	100.0%	High	607	1	213	quarantine
11:30:52	simulation	simulation	Attack	100.0%	Critical	29	1	1,498	quarantine
11:30:48	simulation	simulation	Attack	100.0%	High	11	1	1,911	quarantine
11:29:43	simulation	simulation	Benign	77.5%	Low	3	4	200	monitor
11:29:40	simulation	simulation	Benign	56.5%	Low	3	4	100	monitor
11:29:37	simulation	simulation	Attack	100.0%	High	607	1	213	quarantine
11:29:34	simulation	simulation	Attack	100.0%	Critical	29	1	1,498	quarantine
11:29:31	simulation	simulation	Attack	100.0%	High	11	1	1,911	quarantine

Threat Detail

Attack @ 100.0% confidence · severity **Critical** (85)

Measured traffic: 1,592 pkt/s, 464,059 bytes/s, 37 sources, 8 destinations.

Distributed-traffic explanation: Observed 37 unique sources and 8 unique destinations at 1592 packets/sec (464059 bytes/sec); traffic is distributed across many sources and destinations. (score 0.7603, High).

Severity factors:
- Random Forest classified the window as Attack (100.0% confidence)
- High model confidence (>= 90%)
- Elevated packet rate (1592 pkt/s)
- High distributed-traffic risk (0.76): Observed 37 unique sources and 8 unique destinations at 1592 packets/sec (464059 bytes/sec); traffic is distributed across many sources and destinations.
- Wide traffic spread (37 sources, 8 destinations)

Recommended mitigation: Quarantine top sources in SentinelFlow immediately and rate-limit the affected interface.

Mitigation status: quarantine · **Recovery:** quarantines auto-expire (default ~40 s).

Model features: `average_received_packet_size=291.5, average_sent_packet_size=0, received_bytes=1,392,177, received_bytes_per_second=464,059, received_packets=4,776, received_packets_per_second=1,592, sent_bytes=0, sent_bytes_per_second=0, sent_packets=0, sent_packets_per_second=0`

Mitigation Center

Active quarantines (auto-recover)

- **10.66.3.234** — SentinelFlow Attack window (simulation) [High] (4s left)
- **10.66.27.109** — SentinelFlow Attack window (simulation) [High] (4s left)
- **10.66.4.63** — SentinelFlow Attack window (simulation) [High] (4s left)
- **10.66.36.189** — SentinelFlow Attack window (simulation) [Critical] (47s left)
- **10.66.6.158** — SentinelFlow Attack window (simulation) [Critical] (47s left)
- **10.66.32.204** — SentinelFlow Attack window (simulation) [Critical] (47s left)
- **10.99.20.125** — SentinelFlow Attack window (simulation) [High] (10s left)
- **10.99.7.194** — SentinelFlow Attack window (simulation) [High] (10s left)
- **10.99.19.131** — SentinelFlow Attack window (simulation) [High] (10s left)
- **10.99.19.118** — SentinelFlow Attack window (simulation) [High] (20s left)
- **10.99.13.142** — SentinelFlow Attack window (simulation) [High] (20s left)
- **10.99.8.144** — SentinelFlow Attack window (simulation) [High] (20s left)
- **10.66.25.97** — SentinelFlow Attack window (simulation) [Critical] (63s left)
- **10.66.12.205** — SentinelFlow Attack window (simulation) [Critical] (63s left)
- **10.66.17.111** — SentinelFlow Attack window (simulation) [Critical] (63s left)
- **172.16.37.195** — SentinelFlow Attack window (simulation) [Critical] (78s left)
- **172.16.2.26** — SentinelFlow Attack window (simulation) [Critical] (78s left)
- **172.16.27.38** — SentinelFlow Attack window (simulation) [Critical] (78s left)

History

11:31:22 · 172.16.27.38 · active · SentinelFlow Attack window (simulation) [Critical]
11:31:22 · 172.16.2.26 · active · SentinelFlow Attack window (simulation) [Critical]
11:31:22 · 172.16.37.195 · active · SentinelFlow Attack window (simulation) [Critical]
11:31:07 · 10.66.17.111 · active · SentinelFlow Attack window (simulation) [Critical]